Spectral Theory: Applications

Problem

Alice always eats lunch either at restaurant A or restaurant B. Suppose that

- *Initially she is equally likely to eat at each restaurant.* $\longrightarrow \vec{S}_0 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$
- *She never eats at A two days in a row.*
- *If she eats at restaurant B one day, then the next day she is three times as likely to eat at B as at A.*

Calculate the probabilities of her eating at A and B for the first week. What do you notice?

$$\text{(transition matrix)} \quad T = \begin{array}{c} \overbrace{\begin{array}{cc} A & B \end{array}}^{\text{today}} \\ \begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix} \end{array} \left.\begin{array}{c} A \\ B \end{array}\right\} \text{next day}$$

$$\vec{S}_1 = T\vec{S}_0 = \begin{bmatrix} 0.125 \\ 0.875 \end{bmatrix}$$

$$\vec{S}_2 = T\vec{S}_1 = \begin{bmatrix} 0.2188 \\ 0.7812 \end{bmatrix}$$

$$\vdots$$

$$\vec{S}_6 = T\vec{S}_5 = \begin{bmatrix} 0.2001 \\ 0.7999 \end{bmatrix}$$

$$\vec{S}_7 = T\vec{S}_6 = \begin{bmatrix} 0.2001 \\ 0.7999 \end{bmatrix}$$

The state vectors seem to be converging to $\begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$.

Also, we tried changing the initial state vector and it seemed that the transition matrix above seems to cause the state vectors to converge to $\begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$ independent of the choice of $\vec{S}_0$.

Steady state

Sometimes, as in the example of the customer who ate lunch in restaurant A or B, the state vectors converge to a particular vector, called the steady state vector.

Questions:

- How do we know if a Markov chain has a steady state vector?
- If a Markov chain has a steady state vector, how do we find it?

First, need to learn about regular and stochastic matrices.

Definition

The matrix P is a regular matrix, if for some integer $k > 0$, all entries of P^k are strictly positive.

Example: $T = \begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix}$ is a regular matrix since $T^2 = \begin{bmatrix} \frac{1}{4} & \frac{3}{16} \\ \frac{3}{4} & \frac{13}{16} \end{bmatrix}$ and all the entries of T are positive.

Example: $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is not a regular, since $P^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $P^3 = P$, $P^4 = I$, ... In other words,

$P^n = \begin{cases} P & \text{odd } n \\ I & \text{even } n \end{cases}$, hence P^n always has zero as its entries, which means

Definition

A square matrix P is a stochastic matrix, if its columns are probability vectors.

As an example, consider a 2x2 stochastic matrix whose entries are all nonzero.

$$T = \begin{bmatrix} 1-p & q \\ p & 1-q \end{bmatrix} \quad \text{where} \quad 0 < p, q < 1$$

Problem

Show that stochastic matrices (as defined above) preserve the sum of entries when acting on vectors in $\mathbb{R}^n$.

Assume T is a stochastic matrix and $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$, then

$$\text{ith-entry of } T\vec{x} = (T\vec{x})_i = \sum_{j=1}^{n} T_{ij} x_j$$

$$\text{Sum of entries of } T\vec{x} = \sum_{i=1}^{n} \sum_{j=1}^{n} T_{ij} x_j = \sum_{j=1}^{n} \underbrace{\left(\sum_{i=1}^{n} T_{ij} \right)}_{\substack{\text{equal to one} \\ \text{since } T \text{ is stochastic.}}} x_j = \sum_{j=1}^{n} x_j = \text{Sum of entries of } \vec{x}$$

5. Problem: Show that every stochastic matrix has eigenvalue one.

(I) Note that $\vec{x} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$ is an eigenstate of A^T with eigenvalue one. For example:

$$A = \begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix} \qquad A^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Note that each entry of $A^T \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is equal to the sum of entries in a column of A and since columns of A are probability vectors, they add up to one.

(II) Also, $\det(A - \lambda I) = \det\left((A - \lambda I)^T\right) = \det(A^T - \lambda I)$, which means A and A^T have the same eigenvalues.

(I), (II): A has eigenvalue one.

6. Problem: Show that eigenvalues of a stochastic matrix A satisfy $|\lambda| \leqslant 1$.

Assume $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ be an eigenvector of A^T with eigenvalue λ.

Assume that x_i is the entry of $\vec{x}$ with largest absolute value: $|x_j| \leqslant |x_i|$ for all j.

$$|\lambda x_i| = \left| \sum_j a_{ji} x_j \right| \leqslant \sum_j |a_{ji} x_j| = \sum_j a_{ji} |x_j| \leqslant \underbrace{\sum_j a_{ji}}_{=1 \text{ (Since } A \text{ is stochastic)}} |x_i| = |x_i|$$

Since $\lambda \vec{x} = A^T \vec{x}$ and we are focusing on the ith entry of $\lambda \vec{x}$, where $A = [a_{ij}]$ and $A^T = [a_{ji}]$.

Absolute value of sum is smaller than or equal to sum of abs. values.
$\left|\frac{1}{2} - \frac{1}{2}\right| \leqslant \left|\frac{1}{2}\right| + \left|-\frac{1}{2}\right|$

$|\lambda x_i| \leqslant |x_i|$
$|\lambda| \leqslant 1$

Finaly, since A and A^T have the same eigenvalues, we have completed the proof.